

GRUHESH SRI SAI KARTHIK KURRA

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AI/ML research engineer who builds systems from the math up — transformers, diffusion, and attention re-implemented from scratch, then shipped to production and on-device. Published in IEEE ICCMMLA 2025; independent research on neural weight compression and deepfake forensics. B.Tech CSE, 9.72/10 CGPA.

EDUCATION

KL University

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.72/10

• English proficiency: IELTS Academic 7.5 · Cambridge C1 Advanced.

Hyderabad, India

Aug 2021 – May 2025

EXPERIENCE

AI Integration Engineer

Oct 2025 – Nov 2025

InfoAjax Consulting Pvt Ltd

Remote

- Led development of an AI-driven integration framework connecting enterprise applications with cybersecurity services through standardized APIs and intelligent orchestration.
- Architected an agent-based system using Azure OpenAI for automated schema mapping and adaptive connection strategies that learn from historical patterns.
- Built a full-stack app (React 18, Vite, TailwindCSS, FastAPI) with WebSocket integration for real-time monitoring and agent coordination.
- Integrated Azure Active Directory via Microsoft Graph API for application discovery and automated metadata management.

WSO2 API Developer

Oct 2024 – Jun 2025

InfoAjax Consulting Pvt Ltd

Remote

- Developed and deployed RESTful APIs for PLDT (Philippines' largest telecom) using WSO2 Integration Studio for ticketing-system integration.
- Managed production ticket systems with secure credential handling via GitLab and Choreo cloud hosting with vault-based secrets management.
- Debugged systematically in an environment with sparse WSO2 documentation; engaged international clients directly under strict SLAs.

Founder & CEO

Mar 2024 – Jan 2025

Zynthetix — Personal Research Venture

Remote

- Designed and built a synthetic-data generation platform producing privacy-preserving tabular, image, and text datasets with GANs and VAEs to address dataset scarcity.
- Prototyped a hierarchical meta-controller that coordinates specialized generator models to produce large, de-duplicated datasets, paired with a transformer-based PII detection and masking layer.
- Led a 3-developer team; established Git workflows and CI/CD.

Research Intern

Dec 2024 – Jun 2025

IIIT Hyderabad

Hyderabad, India

- Trained in LLM optimization — quantization, pruning, and knowledge distillation.
- Led a Layout-Preserving Document Translation system with a Vision-Transformer layout detector (92% accuracy) and a FastAPI backend integrating Tesseract OCR for multi-language extraction.
- Presented at the IIIT Hyderabad research expo (May 2025) with live demos to faculty and industry visitors.

AI & ML Research Trainee

May 2024 – Oct 2024

iHub-Data, IIIT Hyderabad

Hyderabad, India

- Completed a 6-month apprenticeship in AI/ML research methodology: LLM fundamentals, prompt engineering, fine-tuning, quantization, and deployment.

Data Science Intern

Jul 2023 – Oct 2023

Xtraleap India Pvt Ltd

Remote

- Built end-to-end ML pipelines — EDA, statistical modeling, data preprocessing, feature engineering, and model evaluation with pandas, NumPy, and scikit-learn.

PROJECTS

MANCE: Morphology-Aware Nested Character Embeddings | *PyTorch, NLP*

[Code](#)

- Word-representation framework modeling words as nested character sequences (n-grams through full words), fused hierarchically with character-level Transformers, CNNs, and BiLSTMs to capture morphology that word-level embeddings miss.
- +58% accuracy on morphologically-varied sentiment (91.67% vs 33.33% baseline), 99.0% on DBpedia, and outsized gains on German NER (GermEval 2014); beats FastText on plurals, past-tense, and superlatives. Accepted at IEEE ICCCLMA 2025.

DeepGuard — iOS On-Device Deepfake Detector | *Swift, Core ML, FastAPI*

[Code](#) | [Demo](#)

- EfficientNet-B1 exported to a 26MB Core ML model running fully on-device at <500ms inference; 98.62% accuracy on 25k images, trained in 55 min on an M4.

BIE: Bit-Index Encoding for Weight Compression | *PyTorch, NumPy, Numba*

[Code](#) | [Zenodo](#)

- 40× compression at 95% sparsity with MSE < 10⁻⁶; three encoding variants (standard, run-length, hybrid) JIT-optimized with Numba, benchmarked across 253 configurations.

Hybrid RAG-Enhanced Deepfake Detection | *RAG, FAISS, CNN*

[Zenodo](#)

- Privacy-first detector combining RAG with visual anomaly detection; 94.8% accuracy, 94.6% F1. Two-stage design with a FAISS vector DB and a custom CNN for facial/lighting/temporal artifacts.

Large-Scale Models From Scratch — GPT-2 & NL2SQL | *PyTorch*

[GPT-2](#) | [NL2SQL](#)

- Implemented the complete GPT-2 (124M parameters) architecture from scratch — custom BPE tokenizer, multi-head attention, and full training pipeline with LR scheduling and gradient accumulation.
- Seq2seq NL→SQL with attention supporting joins, aggregations, and nested subqueries; perplexity 1.42.

Clinical Trial Similarity Analysis | *PubMedBERT, FAISS*

[Code](#)

- Semantic grouping with PubMedBERT embeddings — 99.5% similarity accuracy, 0.891 top-10 retrieval precision across 10,000+ trial descriptions with FAISS indexing.

From-Scratch Collection — 25+ algorithms rebuilt from the math | *NumPy, PyTorch*

[GitHub](#)

- Transformers, Diffusion (DDPM), GANs, VAEs, attention mechanisms; GNNs (GCN 79.3%, GraphSAGE 76.8%, GAT 81.9%); RL (TD, SARSA, Q-learning, MDPs); classic ML with no sklearn (SVM/SMO, random forest, t-SNE, UMAP, DBSCAN).

PUBLICATIONS

- **Morphology-Aware Nested Character Embeddings for Word Representation**, IEEE ICCCLMA 2025 (7th). [IEEE Xplore](#)
- **BIE: Bit-Index Encoding**, Zenodo 2025, DOI: 10.5281/zenodo.17217218. [Link](#)
- **Hybrid RAG-Enhanced Deepfake Detection**, Zenodo 2024, DOI: 10.5281/zenodo.16732053. [Link](#)
- **Dynamic Auto-Finetuning of Language Models**, IJNRD vol. 9(11), 2024.
- **Global Remote RAM Sharing Framework**, IJNRD vol. 9(12), 2024.
- **Voice-Activated AI for Seamless Computer Interaction**, IJNRD vol. 10(1), 2025.

TECHNICAL SKILLS

Primary: Python, PyTorch, Hugging Face Transformers, NumPy, Azure OpenAI, FastAPI, Docker, Git

Specializations: LLMs (GPT-2 124M, fine-tuning, prompt engineering, optimization); Generative (GANs, VAEs, Diffusion/DDPM); Transformers & attention; GNNs; model compression (quantization, pruning, bit-index encoding); RAG; Vision/NLP (ViT, OCR, deepfake detection, PubMedBERT, NL2SQL); synthetic data

Secondary: TensorFlow, LangChain, LlamaIndex, scikit-learn, Core ML; React, Next.js, Node.js, Express.js, React Native, TailwindCSS, Vite

Languages: Python, R, JavaScript, TypeScript, Java, C, C++, SQL, PHP, HTML, CSS

Cloud & DevOps: AWS (EC2, S3, Lambda), Azure (AD, OpenAI, Graph API), GCP, GitLab, CI/CD, WSO2

Databases: PostgreSQL, MongoDB, MySQL, Supabase, Firebase, Redis

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (2023)
- AWS Certified Cloud Practitioner (2023)
- Oracle Cloud Infrastructure Architect Associate (2023)
- Google TensorFlow Developer Certificate (2024)
- RedHat Certified Enterprise Application Developer (2024)
- Oracle Cloud Infrastructure Generative AI Professional (2024)
- Automation Anywhere Advanced Automation Professional (2024)
- WSO2 Certified Micro Integrator Practitioner V4 (2025)

AWARDS & LEADERSHIP

- **1st Place — University Webathon**, KL University (2022): built a diet-management platform in a 4-hour hackathon vs 50+ teams.
- **2nd Place — Design Expo**, KL University (2022–23): Arduino-based smart switchboard for IoT home automation with energy monitoring.
- **Technical Lead, Student Activity Council**, KL University (2023–25): mentored 20+ juniors, ran workshops for 100+ students, built the SAC website with automated certificate generation; SAC Momentum Award.
- **National Tennis Ranking 395 (India), AITA** (2014–19): 100+ national/state tournaments.